

page 8 & 4; Ex 132

$$K(t) = 4$$

$$Y = ax^2 - \text{unknown}$$

$$x(t) = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|}$$

$$1) Y = ax^2$$

$$\vec{r} = \begin{cases} x = t \\ y = at^2 \\ z = 0 \end{cases}$$

$$\vec{r}' = \langle 1, 2at, 0 \rangle$$

$$\vec{r}'' = \langle 0, 2a, 0 \rangle$$

$$2) x=0, y=0, \text{ because there } t=0$$

$$3) \vec{r}' \times \vec{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2at & 0 \\ 0 & 2a & 0 \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} \\ 1 & 2at \\ 0 & 2a \end{vmatrix} = 2ak$$

$$|\vec{r}' \times \vec{r}''| = 2a$$

$$|\vec{r}'| = \sqrt{1^2 + (2at)^2} = \sqrt{1 + 4a^2t^2}$$

$$K(0) = \frac{2a}{\sqrt{1}} = 4 \Rightarrow 2a = 4, a = 2$$

Answer: $y = 2x^2$

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$$x = \sqrt{t^2 + 3}$$

$$y = \ln(t^2 + 3)$$

$$z = t$$

$$(2, \ln 4, 1)$$

$$1) \vec{r}(t) = \langle \sqrt{t^2 + 3}, \ln(t^2 + 3), t \rangle$$

$$\vec{r}'(t) = \left\langle \frac{3t}{2\sqrt{t^2 + 3}}, \frac{2t}{t^2 + 3}, 1 \right\rangle$$

$$2) z = t \quad (2, \ln 4, 1) \text{ so } z = 1$$

$$t = 1$$

$$3) \vec{r}(1) = \langle 2, \ln 4, 1 \rangle \Rightarrow x_0, y_0, z_0$$

$$\vec{r}'(1) = \left\langle \frac{3}{4}, \frac{1}{2}, 1 \right\rangle \Rightarrow a, b, c$$

$$4) x = x_0 + at$$

$$y = y_0 + bt \quad z = z_0 + ct$$

$$x = 2 + \frac{3}{4}t$$

$$y = \ln 4 + \frac{1}{2}t \quad z = 1 + t$$

answer:

$$f(t) = \begin{cases} x = 2 + \frac{3}{4}t \\ y = \ln 4 + \frac{1}{2}t \\ z = 1 + t \end{cases}$$

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$$Z = x^2 - y^2$$

$$x^2 + y^2 = 1$$

$$x^2 + y^2 = 1$$

$$x^2 + y^2 = r^2 (r=1)$$

~~$$x = r \cos \theta \Rightarrow x = \cos \alpha$$~~

~~$$y = r \sin \alpha \Rightarrow y = \sin \alpha$$~~

~~$$x = r \cos \alpha \Rightarrow x = \cos \alpha$$~~

$$y = r \sin \alpha = \sin \alpha$$

$$Z = \cos^2 \alpha - \sin^2 \alpha = \cos 2\alpha$$

$$\vec{r} = \langle \cos \alpha, \sin \alpha, \cos 2\alpha \rangle$$

$$0 \leq \alpha \leq 2\pi$$